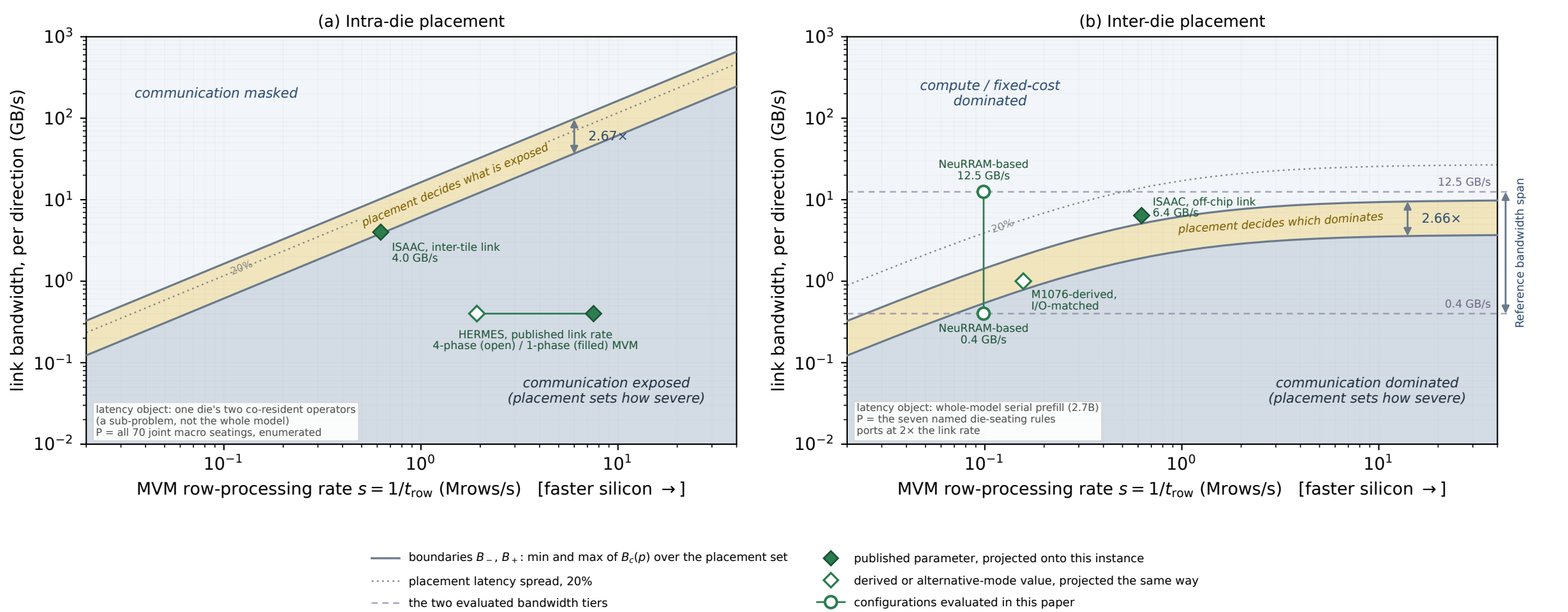
(a) Intra-die placement
communication masked
placement decides what is exposed
2.67×
20%
ISAAC, inter-tile link
4.0 GB/s
HERMES, published link rate
4-phase (open) / 1-phase (filled) MVM
communication exposed
(placement sets how severe)
latency object: one die's two co-resident operators
(a sub-problem, not the whole model)
P = all 70 joint macro seatings, enumerated
link bandwidth, per direction (GB/s)
MVM row-processing rate $s = 1/t_{row}$ (Mrows/s) [faster silicon →]
(b) Inter-die placement
compute / fixed-cost
dominated
NeuRRAM-based
12.5 GB/s
12.5 GB/s
ISAAC, off-chip link
6.4 GB/s
placement decides which dominates
2.66×
20%
M1076-derived,
I/O-matched
0.4 GB/s
NeuRRAM-based
0.4 GB/s
Reference bandwidth span
communication dominated
(placement sets how severe)
latency object: whole-model serial prefill (2.7B)
P = the seven named die-seating rules
ports at 2× the link rate
link bandwidth, per direction (GB/s)
MVM row-processing rate $s = 1/t_{row}$ (Mrows/s) [faster silicon →]
boundaries B_-, B_+: min and max of $B_c(p)$ over the placement set
placement latency spread, 20%
the two evaluated bandwidth tiers
published parameter, projected onto this instance
derived or alternative-mode value, projected the same way
configurations evaluated in this paper